

WHAT WILL WE COVER?

We start with your first Python commands, then build up to storing, checking, and converting values.

1

print() and comments

Display output and annotate your code

2

Variables

Store values and name them

3

Data types

str, int, float, bool — and how to check them

4

Type conversion and input

Convert between types and read from the keyboard

Goal: write a program that stores personal information in variables and prints a formatted profile.

PRINT()

`print()` tells Python to display something on the screen. It is the first command you will use in almost every program.

WHAT IT DOES

Displays output

Whatever you put inside the parentheses appears on the screen when the program runs. Python processes each `print()` line in order, top to bottom.

You can print text, numbers, variables, or a mix of all three.

SYNTAX

Text needs quotes

```
print("Hello, World!")
print("My name is Sara")
print(42)
print(3.14)
print("I am", 25, "years old")
```



Use **double** quotes `"..."` or **single** quotes `'...'` around text — both work. Numbers go in without quotes.

PYTHON BASICS

PRINT() IN ACTION

```
print("Hello, World!")  
print("My name is Sara")  
print("I am", 25, "years old")
```

```
# Print a blank line  
print()
```

```
# Numbers – no quotes needed  
print(42)  
print(3.14)
```



OUTPUT

```
Hello, World!  
My name is Sara  
I am 25 years old
```

```
42  
3.14
```



COMMENTS

A comment is a note written for the programmer. Python ignores everything after a `#` symbol on that line.

WHY USE THEM

Explain your thinking

Comments do not affect what the program does. They help you and others understand *why* the code is written the way it is.

Professional code has comments. Get in the habit early.

SYNTAX

Start with `#`

```
# This whole line is a comment
print("Hello!") # comment at end of line

# Describe what the next line does
print("My name is Sara")

# You can comment out code to disable it
# print("This line will not run")
```



VARIABLES

WHAT IS A VARIABLE?

A **variable** is a named box that stores a value. You give it a name, and Python remembers the value for you.

WITHOUT A VARIABLE

```
print("Sara")  
print("Sara")  
print("Sara")
```



<p class="var-compare-note">You repeat the value every time. If it changes, you must up

WITH A VARIABLE

```
name = "Sara"  
  
print(name)  
print(name)  
print(name)
```



<p class="var-compare-note">Change the value in one place – it updates everywhere automa

VARIABLES

CREATING VARIABLES

Write the name, then `=`, then the value. Python creates the variable and remembers the value.

EXAMPLES

All four types

```
name      = "Sara"    # text
age       = 25        # whole number
height    = 1.68      # decimal number
is_student = True     # True or False

# Use them
print(name)          # Sara
print(age)            # 25

# Update a value any time
age = 26
print(age)            # 26
```



NAMING RULES

Valid names

- ✓ Use letters, digits, underscores
- ✓ Start with a letter or underscore
- ✓ Use `snake_case` for readability
- ✗ Cannot start with a digit — `1name`
- ✗ No spaces — use `my_name` not `my name`
- ✗ Case matters — `name` $\neq$ `Name`

DATA TYPES

THE FOUR BASIC TYPES

Every value in Python has a **type**. The type tells Python what the value is and what you can do with it.

STR

String

Text — always inside quotes.

```
name = "Sara"  
city = 'Berlin'  
email = "sara@redi.de"
```



INT

Integer

Whole numbers — no quotes, no decimal point.

DATA TYPES

TYPE() AND LEN()

```
name    = "Sara"
age     = 25
height  = 1.68
flag    = True

# type() tells you the type of a value
print(type(name))    # <class 'str'>
print(type(age))     # <class 'int'>
print(type(height))  # <class 'float'>
print(type(flag))    # <class 'bool'>

# len() counts characters in a string
print(len(name))     # 4
print(len("Berlin")) # 6
print(len(""))       # 0 - empty string
```



WHY USE THEM

type()

TYPE CONVERSION

CONVERTING BETWEEN TYPES

```
# str → int
age = int("25")
print(age + 1)          # 26

# str → float
price = float("9.99")
print(price * 2)        # 19.98

# int/float → str (needed when joining text)
age = 25
print("I am " + str(age) + " years old")

# int → float and back
x = float(5)           # 5.0
y = int(3.9)           # 3 ← decimal is cut, not rounded!
```



THE THREE CONVERTERS

`int()` Converts to a whole number. Cuts off any decimal — does not round.

`float()` Converts to a decimal number.

TYPE CONVERSION

COMMON TYPE ERRORS

TYPEERROR

Cannot mix str and int

```
# x This crashes
age = 25
print("I am " + age)
```



Fix: convert with `str(age)` or use a comma instead of `+`

```
print("I am " + str(age)) # ✓
print("I am", age)       # ✓ comma works too
```



VALUEERROR

Cannot convert non-numbers

```
# x This crashes
age = int("twenty five")
```

USER INPUT

INPUT()

```
# Ask for text – always returns a string
name = input("What is your name? ")
print("Hello, " + name + "!")

# Ask for a number – must convert!
age = int(input("How old are you? "))
print("Next year you will be", age + 1)

# Ask for a decimal number
price = float(input("Enter a price: "))
print("Double:", price * 2)
```



WHAT HAPPENS ```text What is your name? Sara Hello, Sara! How old are you? 25 Next year you will be 26 Enter a price: 4.99 Double: 9.98 ```

`input()` **always** returns a string — even when the user types a number. Wrap it with `int()` or `float()` if you need to do maths.

EXERCISE

BUILD AN ABOUT ME PROGRAM

1 Create variables Name five variables about yourself: `name`, `age`, `city`, `language`, `is_student`.

2 Print each one Use `print()` to display every variable on its own line.

3 Check the types Use `type()` to print the type of each variable.

4 Update your age Change `age` to next year's value and print it again.

5 Measure your name Use `len(name)` to print how many letters your name has.

Done when: your output shows all five values, five types, the updated age, and the name length.

```
# Lesson 2 – About Me

# 1. Create your variables
name      =
age       =
city      =
language  =
is_student =

# 2. Print each variable

# 3. Check types with type()
```

About Me — solution

1. Variables

```
name      = "Sara"  
age       = 25  
city      = "Berlin"  
language  = "Arabic"  
is_student = True
```

2. Print each one

```
print(name)      # Sara  
print(age)       # 25  
print(city)      # Berlin  
print(language)  # Arabic  
print(is_student) # True
```

3. Types

```
print(type(name)) # <class 'str'>  
print(type(age))  # <class 'int'>  
print(type(is_student)) # <class 'bool'>
```

4. Update age

```
age = 26  
print(age)      # 26
```

Type detective

For each code snippet, predict the output before you run it. Then run the code to check.

1 What does `type("42")` return?

3 What does `len("ReDI")` return?

5 What does `str(True)` return?

2 What does `int(3.9)` return?

4 Does `"I am " + 25` work? Why or why not?

6 What does `type(3.0)` return?

Type detective — solution

1

`type("42")`

`<class 'str'>` — quotes make it a string, not a number.

2

`int(3.9)`

`3` — `int()` cuts the decimal off; it does not round up.

3

`len("ReDI")`

`4` — four characters.

4

`"I am " + 25`

No — `TypeError`. Use `str(25)` first, or a comma.

5

`str(True)`

`"True"` — the boolean becomes the string `"True"`.

6

`type(3.0)`

`<class 'float'>` — the decimal point makes it a float.

For next class

- 1 **All About Me** — create variables for `name`, `age`, `city`, `favourite_food`, `favourite_colour`, `is_working`. Print each on its own line.
- 2 **Type inspector** — for every variable above, print its type with `type()`.
- 3 **Receipt** — create `item`, `price` (float), `quantity` (int). Print: `Bread - quantity: 2 - price: 1.5`
- 4 **len() explorer** — create three string variables (name, city, food). Print the length of each. Which is longest?
- 5 **Mad libs** — create `adjective`, `animal`, `place`, `number`. Print: *The [adjective] [animal] visited [number] places in [place].*
- 6 **Converter chain** — start with `age_text = "28"`. Convert it to `int`, then to `float`, then back to `str`. Print the type after each step.
- 7 **Interactive profile** — use `input()` to ask for a name and age. Print a sentence using both. Bonus: print how many letters the name has.

YOU HAVE VARIABLES

1 print() and comments 2 Variables 3 str · int · float · bool 4 type() · len() · input()

Next lesson: arithmetic, order of operations, and building a calculator with user input.